

Ronen Wdowinski

Combinatorics Group – Institute of Discrete Mathematics
Graz University of Technology (TU Graz)
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CURRENT POSITION

Postdoctoral Researcher Sep. 2024 – present
Supervisor: Mihyun Kang
Graz University of Technology (TU Graz), Graz, Austria

EDUCATION

PhD in Combinatorics and Optimization May 2021–Aug. 2024
Supervisor: Penny Haxell
Thesis title: Arboricity and transversal problems on bounded degree graphs
University of Waterloo, Waterloo ON, Canada

Master of Mathematics (Combinatorics and Optimization) Sep. 2019–Apr. 2021
Supervisor: Penny Haxell
University of Waterloo, Waterloo ON, Canada

Bachelor of Science (Major: Mathematics) Aug. 2015–Jun. 2019
Rice University, Houston TX, USA

PREVIOUS EMPLOYMENT

Instructor Jan.–Apr. 2024
University of Waterloo, Faculty of Mathematics
MATH 136: Linear Algebra 1 for Honours

Teaching Assistant Sep. 2019–Dec. 2023
University of Waterloo, Faculty of Mathematics

- MATH 239: Introduction to Combinatorics three terms, 2019–2020
- MATH 249: Introduction to Combinatorics (Advanced) two terms, 2021–2023
- CO 250: Introduction to Optimization two terms, 2020–2021
- CO 331: Coding Theory two terms, 2021–2022
- CO 342: Introduction to Graph Theory one term, 2022
- CO 351: Network Flow Theory one term, 2021
- CO 370: Deterministic OR Models one term, 2023
- CO 442: Graph Theory two terms, 2020–2022

Researcher May–Jul. 2018
University of Texas at Tyler, Tyler TX, USA

- Studied and characterized special inverse semigroups arising from C^* -algebras and path categories.

Research Assistant Jun.–Aug. 2017
University of Miami, Miami FL, USA

- Applied signal processing techniques to the detection of earthquakes along faultlines, programmed in MATLAB.

1. R. Wdowinski, **Tight constructions for reconfigurations of independent transversals**, arXiv:2604.21576 (2026), submitted.
2. A. Geisler, M. Kang, M. Sarantis, R. Wdowinski, **Sampling from the antiferromagnetic Ising model on bipartite, regular expander graphs**, arXiv:2603.02101 (2026), submitted.
3. R. Wdowinski, **Hall's theorem for reconfigurations and higher dimensional topological connectedness**, arXiv:2511.04863 (2025), submitted.
4. A. Geisler, M. Kang, M. Sarantis, R. Wdowinski, **Counting independent sets in percolated graphs via the Ising model**, *Random Structures & Algorithms* 68(1) (2026), e70044.
5. R. Wdowinski, **Bounded degree graphs and hypergraphs with no full rainbow matchings**, *European Journal of Combinatorics* 133 (2026), 104316.
6. P. Haxell, R. Wdowinski, **Constructing graphs with no independent transversals**, *Electronic Journal of Combinatorics* 31(2) (2024), P2–39.
7. S. Cambie, P. Haxell, R. Kang, R. Wdowinski, **A precise condition for independent transversals in bipartite covers**, *SIAM Journal of Discrete Mathematics* 38(2) (2024), 1451–1461.
8. P. Haxell, R. Wdowinski, **Degree criteria and stability for independent transversals**, *Journal of Graph Theory* 106(2) (2024), 352–371.
9. R. Wdowinski, **On an f -coloring generalization of linear arboricity of multigraphs**, *Discrete Mathematics* 347(2) (2024), 113777.
10. R. Wdowinski, **Orientation-based edge-colorings and linear arboricity of multigraphs**, *Journal of Graph Theory* 102(4) (2023), 633–647.
11. A. Donsig, J. Gensler, H. King, D. Milan, R. Wdowinski, **On zigzag maps and the path category of an inverse semigroup**, *Semigroup Forum* 100 (2020), 790–805.
12. E. Blaisdell, A. Gyárfás, R. A. Krueger, R. Wdowinski, **Partitioning the power set of $[n]$ into C_k -free parts**, *Electronic Journal of Combinatorics* 26(3) (2019), P3–38.

PRESENTATIONS

Conferences and Workshops

1. *Counting independent sets in percolated graphs via the Ising model*. SFB Winter Status Workshop, Graz, Austria, Feb. 2026.
2. *Hall's theorem for reconfigurations via colorful simplices*. Discrete Mathematics Meeting, Graz, Austria, Sep. 2025.
3. *Counting independent sets in percolated graphs via the Ising model*. Random Structures & Algorithms (RSA), Vienna, Austria, Aug. 2025.
4. *Degree conditions for independent transversals*. SFB Kick-off Event, Payerbach, Austria, Sep. 2024.
5. *Bounded degree (hyper)graphs with no transversals*. Fulkerson 100, Waterloo ON, Canada, Jul. 2024.
6. *Bounded degree hypergraphs with no full rainbow structures*. SIAM Conference on Discrete Mathematics (DM24), Spokane WA, USA, Jul. 2024.
7. *Linear arboricity and its f -coloring generalization via orientations*. CanaDAM 2023, Winnipeg MB, Canada, Jun. 2023.
8. *The linear arboricity of sparse multigraphs via orientations*. 26th Ontario Combinatorics Workshop, Waterloo ON, Canada, May 2022.
9. *Characterization of zigzag inverse semigroups*. Mathfest 2018, Denver CO, USA, Aug. 2018.

Seminar presentations

1. *Reconfigurability and topological variations of Hall's theorem*. PhaseCAP seminar at University of Amsterdam. Amsterdam, the Netherlands, Apr. 2026.
2. *Reconfigurability and topological variations of Hall's theorem*. Seminar on Foundations of Computing at Masaryk University. Brno, Czech Republic, Mar. 2026.

3. *Degree conditions for independent transversals*. Advanced Topics in Discrete Mathematics seminar at TU Graz. Graz, Austria, Mar. 2025.
4. *Degree conditions for independent transversals*. Combinatorics, Geometry and Topology seminar at ISTA. Klosterneuburg, Austria, Feb. 2025.
5. *Constructing graphs with no independent transversals*. Graphs and Matroids Seminar at University of Waterloo. Waterloo ON, Canada, Mar. 2024.
6. *An f -coloring generalization of linear arboricity*. Graphs and Matroids Seminar at University of Waterloo. Waterloo ON, Canada, Mar. 2023.
7. *Linear arboricity and its f -coloring generalization via orientations*. Invited talk at the Discrete Math Seminar at Georgia State University. Online, Oct. 2022.
8. *The linear arboricity of sparse multigraphs via orientations*. Graphs and Matroids Seminar at University of Waterloo. Waterloo ON, Canada, Jun. 2022.
9. *Partitioning the power set of $[n]$ into C_k -free parts*. Budapest Semesters in Mathematics. Budapest, Hungary, Dec. 2018.
10. *Higher-order Szegő theorems*. Science in a Flash recruitment event. Rice University, Houston TX, USA, Apr. 2018.
11. *Recurrent behavior in a nonlinear oscillator chain*. Geometry Lab. Rice University, Houston TX, USA, Apr. 2017.

AWARDS

University of Waterloo

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| 1. SIAM Student Travel Award | 2024 |
| 2. Ontario Graduate Scholarship + President's Graduate Scholarship
One of only five recipients for the 2023 Waterloo international competition. | 2023–2024 |
| 3. International Doctoral Student Award | 2021–2024 |
| 4. Sinclair Graduate Scholarship | 2021 |
| 5. Combinatorics and Optimization Graduate Award | 2021, 2023 |
| 6. International Master's Award of Excellence | 2019–2021 |

Rice University

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| 1. Trustee Distinguished Scholarship | 2015–2019 |
| 2. Lynn L. Elsenhans Scholarship for mathematical sciences | 2018–2019 |
| 3. Stephen B. Smith Scholarship (Study abroad scholarship) | 2018 |
| 4. Judge's Choice Award for Science in a Flash (Presentation award) | 2018 |

SERVICE

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| 1. Co-organizer of Graz Combinatorics Seminar at TU Graz | Jan. 2025 – Jun. 2026 |
| 2. Co-organizer of SFB Winter Status Workshop at Graz, Austria | Feb. 2026 |
| 3. Co-organizer of Combinatorics Workshop at Strobl, Austria | Aug. 2025 |
| 4. Co-organizer of SFB reading group on matchings at TU Graz | May 2025 |

ADDITIONAL ACTIVITY: PROGRAMS, SCHOOLS, RESEARCH VISITS, REFEREE

1. *PhaseCAP Workshop on Phase Transitions in Algorithms*. CWI, Amsterdam, the Netherlands, May 2026.
2. *University of Amsterdam Research Visits*. Two one-week visits hosted by Ross Kang. CWI, Amsterdam, the Netherlands, Apr. and May 2026.
3. *SFB Summer School*: planar maps, domino tilings, and Markov chains. Pörttschach am Wörthersee, Austria, Sep. 2025.
4. *IAS/Park City Mathematics Institute (PCMI) Research Program*. Three-week program on extremal and probabilistic combinatorics. Park City UT, USA, Jul. 2025.
5. *ISTA One-Week Research Visit*. Hosted by Benjamin Moore and Matthew Kwan. Klosterneuburg, Austria, Feb. 2025.
6. *DIMEA Combinatorial Potluck 2024*. Workshop with research groups from Brno, Graz, Vienna, and Passau about extremal and probabilistic combinatorics. Brno, Czech Republic, Nov. 2024.
7. *SLMath Summer School*. Concentration inequalities and localization techniques in high dimensional probability and geometry. Berkeley CA, USA, Jul. 2023.
8. *Budapest Semesters in Mathematics*. Undergraduate study abroad, Budapest, Hungary, Sep.–Dec. 2018.
9. *Refereed for*: Proceedings of American Mathematical Society, Journal of Combinatorial Theory Series B, Random Structures & Algorithms, Electronic Journal of Probability, Electronic Journal of Combinatorics, Journal of Graph Theory, Discrete Mathematics, Discrete Applied Mathematics, European Journal of Mathematics.